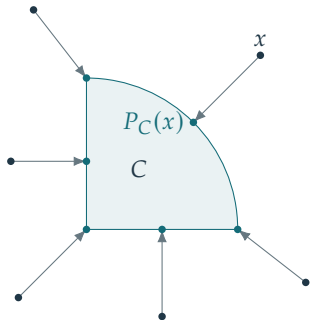
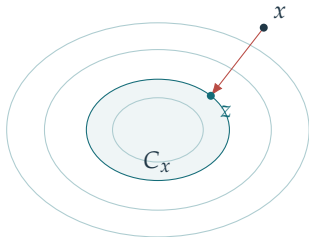


Projection onto a fixed set



$$x - P_C(x) \in N_C(P_C(x))$$

Projection onto an input-dependent sublevel set



$$z = \text{prox}_f(x), \quad C_x = \{u : f(u) \leq f(z)\}$$

$$z = P_{C_x}(x)$$